There are two main reasons. First, on a scale set by atoms, the moon is enormously far away. And, as mentioned, in many circumstances (although by no means all), the quantum equations show that a probability wave typically has an appreciable value in some small region of space and quickly drops nearly to zero as you move away from this region (as in Figure 4.5). So the likelihood that even a single electron that you expect to be in the same room as you—such as one of those that you just exhaled—will be found in a moment or two on the dark side of the moon, while not zero, is extremely small. So small, that it makes the probability that you will marry Nicole Kidman or Antonio Banderas seem enormous by comparison. Second, there are *a lot* of electrons, as well as protons and neutrons, making up the air in your room. The likelihood that *all* of these particles will do what is extremely unlikely even for one is so small that it's hardly worth a moment's thought. It would be like not only marrying your movie-star heartthrob but then also winning every state lottery every week for, well, a length of time that would make the current age of the universe seem a mere cosmic flicker.

This gives some sense of why we do not directly encounter the probabilistic aspects of quantum mechanics in day-to-day life. Nevertheless, because experiments confirm that quantum mechanics does describe fundamental physics, it presents a frontal assault on our basic beliefs as to what constitutes reality. Einstein, in particular, was deeply troubled by the probabilistic character of quantum theory. Physics, he would emphasize again and again, is in the business of determining with certainty what has happened, what is happening, and what will happen in the world around us. Physicists are not bookies, and physics is not the business of calculating odds. But Einstein could not deny that quantum mechanics was enormously successful in explaining and predicting, albeit in a statistical framework, experimental observations of the microworld. And so rather than attempting to show that quantum mechanics was wrong, a task that still looks like a fool's errand in light of its unparalleled successes, Einstein expended much effort on trying to show that quantum mechanics was not the final word on how the universe works. Even though he could not say what it was, Einstein wanted to convince everyone that there was a deeper and less bizarre description of the universe yet to be found.